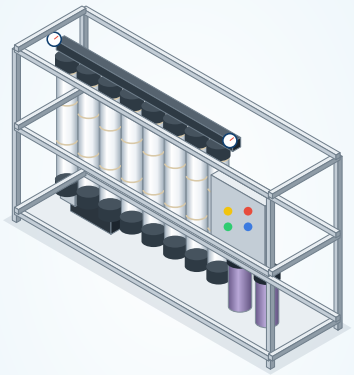
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Safi Treatment systems

Packaged drinking water treatment skids built from ten standard modules. Ultrafiltration (0.01 µm) and UV-C form the core barrier; media filtration, chlorine dosing and salt membranes are added according to the raw water. Stainless steel frame, single or three-phase power, optional solar.



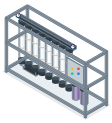
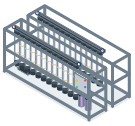
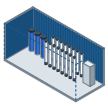
Safi M · schematic view

1 Design basis — raw water at inlet

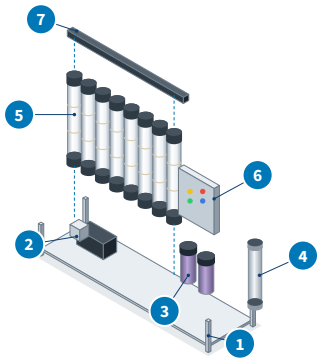
Parameter	Standard line	Above this
Turbidity	≤ 5 NTU	5–50: media pre-filter before M1; > 50: settling
Iron / manganese	≤ 0.3 / 0.08 mg/L	add M2 iron media with chlorine pre-dose
Fluoride	≤ 1.5 mg/L	add M7 fluoride media
TDS	≤ 1,000 mg/L	add M8 NF / RO
pH	6.5–8.5	pH correction by design
Temperature	5–40 °C	—
Supply	from storage tank	—

Designed for the worst value of each parameter in the water test. Feed pump included.

2 Models

Parameter · unit	Safi S	Safi M	Safi L	Containerised
Treated water · m^3/h	0.5–1.0	2.0–4.0	10–20	to design
Daily output (10 h) · m^3/d	5–10	20–40	100–200	to design
UF modules · no.	1–2	4–8	20–40	to design
UF pore size · µm	0.01	0.01	0.01	0.01
UF recovery · %	90–95	90–95	90–95	90–95
Pre-filtration · µm	5 + 1	5 + 1	5 + 1	5 + 1
UV-C · W	sized to flow	55	multiple chambers	to design
Feed pump · kW	0.75–1.1	1.5–2.2	5.5–11	to design
Installed power · kW	≈ 1.5	2.2–3.0	7.5–15	to design
Supply · V / Hz	230 / 50, 1-ph	230 / 50, 1-ph	400 / 50, 3-ph	to design
Specific energy · kWh/m^3	—	≈ 0.75	—	—
Dry weight · kg	80–130	220–250	600–1,200	—



2.1 Inside a Safi M

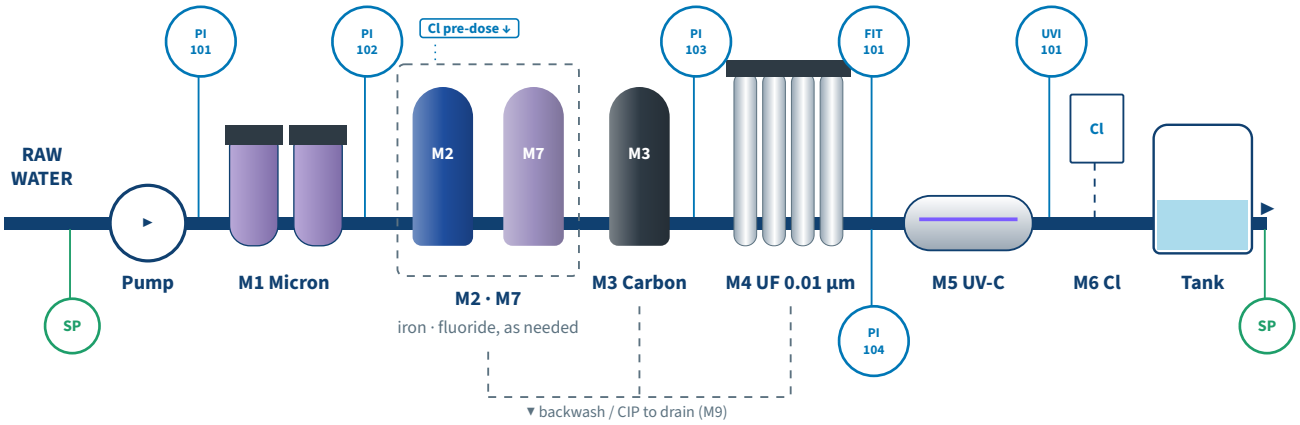
1. Stainless steel skid
2. Feed pump
3. Micron filters (M1)
4. UV-C chamber (M5)
5. UF membranes (M4)
6. Smart Controller (M10)
7. Feed / product manifold

Media vessels (M2, M3, M7) stand beside the skid when the water needs them. Same layout for every size.

Safi M values from the running Kampala unit: rated 4,000 L/h on town water, fitted with M1 + M2 + M3 + M4 + M5 (energy = rated power ÷ rated flow). Safi S and L values indicative, confirmed per design. Recovery is the UF design target. Dimensions supplied per project.

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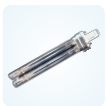
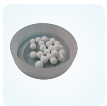
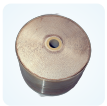
3 Process and instrumentation



PI pressure · **FIT** flow · **UVI** UV intensity · **SP** sample point · dashed box = fitted when the water needs it

Simplified process flow diagram; full P&ID with ISA 5.1 tags supplied per project. UF pressure is read across PI 103 and PI 104. For saline water, M8 NF / RO replaces M4 after the carbon stage. M10 controls the pump, backwash, CIP and alarms.

4 Treatment modules

	Module	Technology	Removes	Service
	**M1** Micron filters	Cartridges 5 µm then 1 µm	Sand, silt, rust	3 months or ΔP > 1 bar
	**M2** Iron & manganese	FRP pressure vessel with catalytic MnO_2 media (e.g. DMI-65), own chlorine pre-dose pump, auto backwash	Dissolved Fe and Mn	Media inspected yearly
	**M3** Activated carbon	Granular activated carbon, auto backwash	Chlorine, taste, odour, colour	Replace 12 months
	**M4** Ultrafiltration	Pressurised hollow-fibre modules in parallel, 0.01 µm, PVDF or PES, 2–4 bar	Bacteria, turbidity, colloids; minerals kept	CIP when pressure rises; membranes 3–5 years
	**M5** UV-C	254 nm lamp, stainless chamber, quartz sleeve	Bacteria, viruses (inactivation)	Lamp 9,000 h; sleeve 24 months
	**M6** Chlorine dosing	Metering pump, sodium hypochlorite	Re-growth in tanks and pipes	Residual 0.2–0.5 mg/L
	**M7** Fluoride media	Activated alumina or bone char	Fluoride	Sized by test; regenerate or replace
	**M8** NF / RO	Spiral-wound membranes, antiscalant dosing	Dissolved salts	Saline water only
	**M9** Cleaning	Flush and backwash valves on the UF manifold; media backwash; CIP set	Fouling of media and membranes	Per controller schedule
	**M10** Smart Controller	PLC, keypad and display, remote data	Runs and protects the line	Checked each service visit

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5 Water categories and module selection

Category	1 Clear town water	2 Ground boreholes, wells	3 Surface rivers, lakes	4 Fluoride Rift Valley	5 Saline brackish
DECIDING TEST VALUES					
Turbidity · NTU	< 5	< 5	5–50	< 5	< 5
Fe / Mn · mg/L	< 0.3 / 0.08	> 0.3 / 0.08	< 0.3 / 0.08	any	any
Fluoride · mg/L	< 1.5	< 1.5	< 1.5	> 1.5	any
TDS · mg/L	< 1,000	< 1,000	< 1,000	< 1,000	> 1,000
MODULES					
M1 Micron filters	●	●	●	●	●
M2 Iron media	○	●	–	○	○
M3 Carbon	●	●	●	●	●
M4 UF	●	●	●	●	○
M5 UV-C	●	●	●	●	●
M6 Chlorine dosing	○	○	●	○	○
M7 Fluoride media	–	–	–	●	–
M8 NF / RO	–	–	–	–	●

● standard ○ added when the test or the site needs it M9 on every system · M10 on every new system
The test values decide the category, not the source: a borehole with low iron is treated as Clear.

6 Treated water quality

Parameter	Target	Tested	
E. coli · /100 mL	absent	0	✓
Total coliforms · /100 mL	absent	0	✓
pH	6.5–8.5	7.59	✓
Fluoride · mg/L	≤ 1.5	0.31	✓
Nitrate · mg/L	≤ 50	2.5	✓
Lead · mg/L	≤ 0.01	< 0.01	✓
Turbidity · NTU	< 1	not in 2026 panel	

Targets: WHO Guidelines for Drinking-water Quality; pH from EAS 153. Tested: Safi M, Kampala, accredited laboratory (Chemiphar), 2026; town-water feed. Product water UNBS certified to US EAS 153:2014.

7 Operating limits and service triggers

Condition	Action
Pressure drop across M1 > 1 bar	change cartridges
UF membrane pressure rises 0.5 bar above clean start	run CIP (M9)
UV intensity below alarm set point	clean sleeve / replace lamp
UV lamp at 9,000 h	replace lamp
Free chlorine at tap < 0.2 mg/L	adjust M6 dose
Feed outside design basis	stop and re-test

INCLUDED

Skid with selected modules · feed pump · controls · commissioning · operator training · first accredited water test.

BY THE CLIENT

Shelter and plinth · raw water supply to inlet · power to panel · storage tanks (unless ordered) · drain connection.

TESTING AND ACCEPTANCE

Factory: pressure, leak and run test before dispatch. Site: commissioning run and accredited lab test. Then a sampling plan agreed with the client.

8 To size a system, send us

Water test
turbidity, Fe, Mn, fluoride, TDS, pH, E. coli

Demand
litres per day and peak hours

Site
power available, space, photos

Use
drinking, bottling or community supply

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